FitBuddy: AI Fitness Plan Generator using Gemini Models
Government Arts and Science College , Manur - MSU142
Team Id:6ab21893a614981d6d2c53e1
Team Name:7

Team Leader:Mohamed Mujahith Kasim M, NMID:DD48BB73EA4A64C8B904D2400C9F7D6F
Member 2:Suman M ,NM ID: 0192097D4D7A60988DA49766C01835E0
Member 3:Udhayanithi S ,NMID:52361B3C0E57DDDB4F242E43149A5A47
Member 4:Malai raja A ,NMID:7255103A926613882F2472E0E276D163

Requirement Analysis Phase – FitBuddy

The **Requirement Analysis Phase** identifies what FitBuddy must provide to users and administrators and defines the technical requirements needed to build the application.

1. Functional Requirements

- **User Registration/Input:** The system should collect the user's name, user ID, age, weight, fitness goal, and workout intensity.
- **Personalized Workout Generation:** Generate a structured **7-day workout plan** based on the user's fitness goal and intensity.
- **Nutrition/Recovery Tips:** Provide a concise nutrition or recovery tip based on the selected fitness goal.
- **Feedback-Based Updating:** Allow users to submit feedback and regenerate the workout plan according to their changing requirements.
- **Data Storage:** Store user information and generated plans in an SQLite database using SQLAlchemy.
- **Admin View:** Provide an admin page to view registered users and their original and updated workout plans.

2. Non-Functional Requirements

- **Usability:** The interface should be simple and easy to use.
- **Performance:** AI responses should be generated efficiently for real-time web interaction.
- **Responsiveness:** The frontend should support different screen sizes using responsive CSS.
- **Maintainability:** The application should follow a modular architecture using FastAPI, separate AI modules, templates, and database logic.
- **Security:** Sensitive information such as the Gemini API key should be stored using environment variables rather than directly in the source code.

3. Technology Requirements

- **Backend:** FastAPI

- **AI:** Google Gemini models
- **Frontend:** HTML, CSS, Jinja2
- **Database:** SQLite with SQLAlchemy
- **Programming Language:** Python
- **Server:** Uvicorn
- **Development:** Virtual environment and required Python libraries.

4. AI Requirements

The AI component should be capable of:

- Generating structured 7-day workout plans.
- Providing actionable nutrition tips.
- Updating plans based on user feedback.
- Producing formatted, context-aware responses suitable for the web application.

5. Overall Requirement

The final requirement is to develop a **user-friendly AI fitness application that accepts individual fitness information, generates personalized workout and nutrition guidance, stores the information, and continuously improves the plan through user feedback**.

Refine FitBuddy's requirements

- Define user and admin use cases
- Prioritize the MVP requirements